

RIGEL · SPEC EDITOR

MCP Guide

Let an AI agent build in the Spec Editor: place objects, write, check, attach and fix Luau, make game modes.

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1 Setting it up

The MCP server is `rigel_mcp.py` in this folder (`Documents\RigelScripts\mcp\`). It needs Python 3.9+ and nothing else. It talks to the **Spec Editor** running on the same PC (the editor listens on `127.0.0.1:47650`; loopback only, so nothing outside the PC can reach it).

1. Start the Spec Editor build and join a server (as for normal editing). The bridge starts with the editor.

2. Add the server to your agent:

- **Claude Code:** `claude mcp add rigel -- python`

- **Claude Desktop** (`claude_desktop_config.json`): `"mcpServers": { "rigel": { "command":`

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- Any other MCP client: run the script over stdio.

3. Check: `python rigel_mcp.py --selftest` prints the editor's status (or says the editor isn't running).

The agent acts **as you**: it has exactly your editor permissions (owner locks, levels, what the server allows).

Everything it does is visible to everyone on the server, like your own edits. Undo works on its edits too.

2 How the world works (read this first)

- **Units and axes.** Positions are Unreal centimetres, `[x, y, z]`. The station is a rotating cylinder around the world **Y axis**; "down" points away from the axis. On the main floor near spawn, the floor is around `z = -27000` to `-29500` depending on the district, so **look at nearby objects' Z** (`list_objects`) before placing things. Rotations are `[pitch, yaw, roll]` in degrees.
- **Handles.** Every object has a handle (a GUID like `2368C681-D689-59C7-3922-DAB4DCB2B201`, sometimes with `_n` after it). Every tool that acts on an object takes the handle. Handles can change when an object is rebuilt (a move, attaching a script), so re-list objects after such edits if a handle stops working.
- **Levels.** Placing needs an **open level** (`level_new(name)` or `level_open(name)`). Objects you place belong to it; `level_save` writes `Documents\RigelLevels\<name>.a2level` with every object, its Game data, its scripts and its script wiring, and any game modes. Levels load back exactly.
- **Areas.** The station is split into module areas (gamemode slots). An object belongs to the area whose box contains it. A script's slots can only point at objects **in the same area**. A **game mode** is an area of its own (see *Game modes*).
- **Replication.** Placing, moving, deleting, Game data, scripts: all go through the server and reach every player, Quest headsets included. Details-panel *properties* marked `replicated: false` are local only.

3 The Luau model (the part that trips everyone up)

- A script is attached to **one object** and runs on **every machine** that has the object: the server, your editor, every player's game. Four players = the script runs five-plus times, each with its own variables.
- Anything that changes shared state must happen **once, on the server**. The game's own components are built for this: e.g. `PhysicalComponent.onOverlapByPlayerServer` fires on the server only, `onOverlapByPlayerClient` only on the player's machine. For game modes, `Rigel.*` actions only take effect on the server, so it's safe that every machine calls them.

- **Never spawn balls from a script** (`spawnBall`, `spawnBallWithParameters`): every machine makes its own ball that nobody else can see. Give a ball spawner the game mode role **ball**, and the server spawns/resets it.
- Script lifecycle: the file runs top to bottom when the object is built, then the game calls `BeginPlay()`. Put setup in `BeginPlay`. Attaching or updating a script **rebuilds** the object (a fresh start).
- Output: `log(...)` and `warn(...)` go to each machine's game log as `LogLuau: Display: [Name.luau]: ...`. Errors appear there too. The MCP reads the editor's own log (`script_logs`), which is enough because the editor runs every script as well.

Typed locals are editor properties (the crash trap)

Every top-level `local Name: Type = ...` **becomes a property of the object**. Only these types can be properties:

Type	What it is
<code>number</code> , <code>string</code> , <code>boolean</code>	a value the user sets in Details (the <code>= value</code> is ignored -- it comes from the editor)
<code>SomethingComponent</code>	a slot : a reference to another object's component, wired in the editor

Any other annotation on a top-level local -- a table type `{number}`, `{[string]: boolean}`, a function type `() -> ()`, a union `number | string` -- **crashes the server and every client**. The server strips such annotations before running, but don't write them:

```
local scores: {number} = {}           -- WRONG: crashes (table type)
local scores = {} :: {number}         -- right: a cast on the value, not a typed local
local onDone: () -> () = function() end -- WRONG: function type
local onDone = function() end         -- right
local speed: number = 5               -- a PROPERTY: starts as whatever the editor says (nil if
unset), not 5
local speed = 5                       -- a plain variable: 5
```

Inside functions, annotate freely (`local x: {number} = {}` inside `BeginPlay` is fine).

Slots: references to other objects

```
local Door: PhysicalComponent = nil    -- a slot (typed top-level local of a Component type)
local Button: BasicButtonComponent = nil

function BeginPlay()
    if (Door :: PhysicalComponent?) == nil or (Button :: BasicButtonComponent?) == nil then
        warn("wire Door and Button")    -- unwired slots are nil
        return
    end
    Button.onButtonPressEvent.Listen(function()
        Door:hideLua(); Door:disableCollisionLua()
    end)
end
```

Wire with `set_script_reference(handle=<scripted object>, script="Name", slot="Door", type="PhysicalComponent", target=<door's handle>)`. `find_reference_targets(type)` lists objects that have that component. A slot may point at the object itself (`target = its own handle`).

Syntax of the API

- Methods: `component:methodName(args)` (C++ `SetCupVisibility` becomes `setCupVisibility`).
- Events: **dot**, then `.Listen: Button.onButtonPressEvent.Listen(function() ... end)`.

- Statics / globals: `VRPawn.getPlayerName(id)`, `LuauClock.timeout(2)`, `Vector.new(x, y, z)`.
- Timers: `LuauClock.timeout(sec):andThen(function() ... end)` (a Promise, also `:await()` inside `Promise.new`), `LuauClock.createTimer(everySec, function() ... end)` repeats, `LuauClock.getTime()` seconds.
- Vectors: `+` `-` `*` `/`, `.x` `.y` `.z`.
- Strings: backtick interpolation works: ``score {n}``.
- Look anything up: `luau_api_search("timer")`, `luau_api_class("TimerComponent")`.

The components you'll use most

Component	Methods	Events
BasicButtonComponent (buttons)	<code>luaEnableButton()</code> , <code>luaDisableButton()</code> , <code>setButtonText(s)</code> , <code>setOnCooldown(b)</code>	<code>onButtonPressEvent()</code>
ToggleableComponent (traps, switches, platforms)	<code>luaEnable()</code> , <code>luaDisable()</code>	<code>blueprintOnEnabled()</code> , <code>blueprintOnDisabled()</code> ; field <code>isEnabled</code>
PhysicalComponent (almost any placed piece)	<code>hideLua()</code> , <code>showLua()</code> , <code>disableCollisionLua()</code> , <code>setDefaultCollision()</code> , <code>setTriggerCollision()</code> , <code>hideLuaClientside()</code>	<code>onOverlapByPlayerServer(playerId)</code> , <code>onOverlapByPlayerClientside()</code> , <code>onOverlapByDisc(disc)</code> , <code>onHitByDisc(disc)</code> , <code>onOverlapEndByPlayerSimple()</code>
TimerComponent (Timer)	<code>start(seconds)</code> , <code>startCountUpFromZero()</code> , <code>stopAndResetTimer()</code>	<code>finished()</code>
BallSpawnerComponent (ball spawners)	<code>getSpawnedBall()</code> , <code>resetBall()</code> , <code>smoothResetBall()</code> -- not <code>spawnBall</code>	<code>discWasSpawnedWithDiscArg(disc)</code> , <code>onBallGrounded()</code>
DiscEntity (a ball)	<code>freeze(hideOutline)</code> , <code>unFreeze(b)</code> , <code>setVelocity(v)</code> , <code>getDiscVelocity()</code> , <code>getDiscPosition()</code> , <code>getGravityDirection()</code>	<code>onGoToSleep()</code>
ScoreComponent (the classic Score board)	<code>setTeamPoints(team0based, n)</code> , <code>getTeamPoints(t)</code> , <code>resetScore()</code>	<code>onScored(goalInfo)</code>
DataTableComponent (Score table)	<code>beginTransaction()</code> , <code>setCellText(text, row, col)</code> , <code>endDataTransaction()</code>	
VRPawn (statics)	<code>getLocalPlayerIndex()</code> (-1 on the server), <code>getPlayerName(id)</code> , <code>getTeamIndexById(id)</code> , <code>getPositionById(id)</code> , <code>getPlayerById(id)</code>	

Quests (statics)	<code>event(playerId, eventId),</code> <code>eventNumber(...),</code> <code>complete(playerId, questId)</code> (-1 = local player)	
Gamemode (the area's game mode)	<code>getStringConfigVariable(k),</code> <code>broadcastEventString(name,</code> <code>value), setTeamScore(team0,</code> <code>n), setTeamRoundsWon,</code> <code>startGame(), stopGame()</code>	<code>onConfigChanged</code> (doesn't fire -- poll)

There is no `TextComponent` from scripts (a slot of that type stays nil). Text goes through **templates**: set a Text object's text to `Score: {score}` (tool `set_text`), then any script calls `Gamemode:broadcastEventString("score", "3")` (or `Rigel.setText("score", "3")`): the server fills `{score}` into every Text object using it, for everyone.

4 The safe workflow for scripts

Always, for every script:

1. **Learn the pieces.** `luau_api_class` for each component you'll touch; `luau_examples` for a working example of something similar; `station_scripts` for how the game itself does it.
2. **Write** the source.
3. `check_luau(source)` -- type-checks against the game's API (wrong method names, wrong argument types, nil misuse) and flags the game's traps (crashing typed locals, client-side ball spawns, endless loops). Fix every error. Warnings deserve a look.
4. `attach_script(handle, name, source)` -- refuses if the check fails. It waits a few seconds and returns the **game log lines** the script produced, the editor's own log and any problems the editor shows.
5. **Read what came back.** Your `log()` lines prove it ran. A runtime error looks like `[string "Name.luau"]:12: attempt to index nil with 'luaEnable' (line 12 of your script)`, or for errors inside promises: `Unhandled Promise rejection: ... [string "Name.luau"]:4: <message>`.
6. **Fix and** `update_script(name, source)` (checked again; every object running it gets the new version). `script_logs(mark=...)` shows only newer lines: note `mark` from one call and pass it to the next.
7. Wire slots with `set_script_reference`, then check the log again (wiring rebuilds the object; `BeginPlay` runs again).
8. `level_save` when it works.

Defensive habits that keep scripts from failing on some machines:

- Nil-check every slot before using it (unwired slots are nil on every machine).
- Wrap calls into components you're not sure about in `pcall(function() ... end)`.
- Never loop forever without yielding; use `LuauClock.createTimer / timeout`.
- Keep per-player effects in client-side events and shared effects in server-side ones.
- `VRPawn.getLocalPlayerIndex()` is `-1` on the server and on desktop spectators.

5 Tools reference

Look around: `editor_status` (start here), `list_objects(filter, near, radius, limit)`, `search_palette(query, category)`, `get_properties(handle, sub_path)`, `get_game_data(handle)`,

`station_scripts(name)`, `level_list`, `gamemode_list`, `editor_logs(since, filter)`, `editor_problems`.

Build: `place_object(item, location, rotation, scale)` returns the new object; `move_object(handle, location, rotation, scale)`; `duplicate_object(handle, location)`; `delete_object(handle)`; `select_object(handle)` (shows it to the user); `set_camera(location, pitch, yaw | handle)`; `undo`, `redo`.

Configure: `set_property(handle, path, value)` -- values in the wire encoding `get_properties` shows (numbers, true/false, x,y,z, p,y,r, r,g,b,a, text); `set_game_data(handle, path, kind, value)` for synced Game data (`sd/...` serverData, `gd/...` gameData, `props/<Field>` script properties); `set_text(handle, text)`.

Luau: `luau_api_search`, `luau_api_class`, `luau_guide(topic)`, `luau_examples(name)`, `check_luau(source, gamemode_code)`, `write_script_file(name, source, save_and_update)`, `read_script_file(name)`, `attach_script(handle, name, source)`, `update_script(name, source)`, `remove_script(handle, name)`, `set_script_reference(...)`, `find_reference_targets(type)`, `script_logs(mark, script, errors_only)`.

Levels: `level_new(name)`, `level_open(name)`, `level_save(name?)`, `level_close`, `level_delete(name)`, `level_upload`.

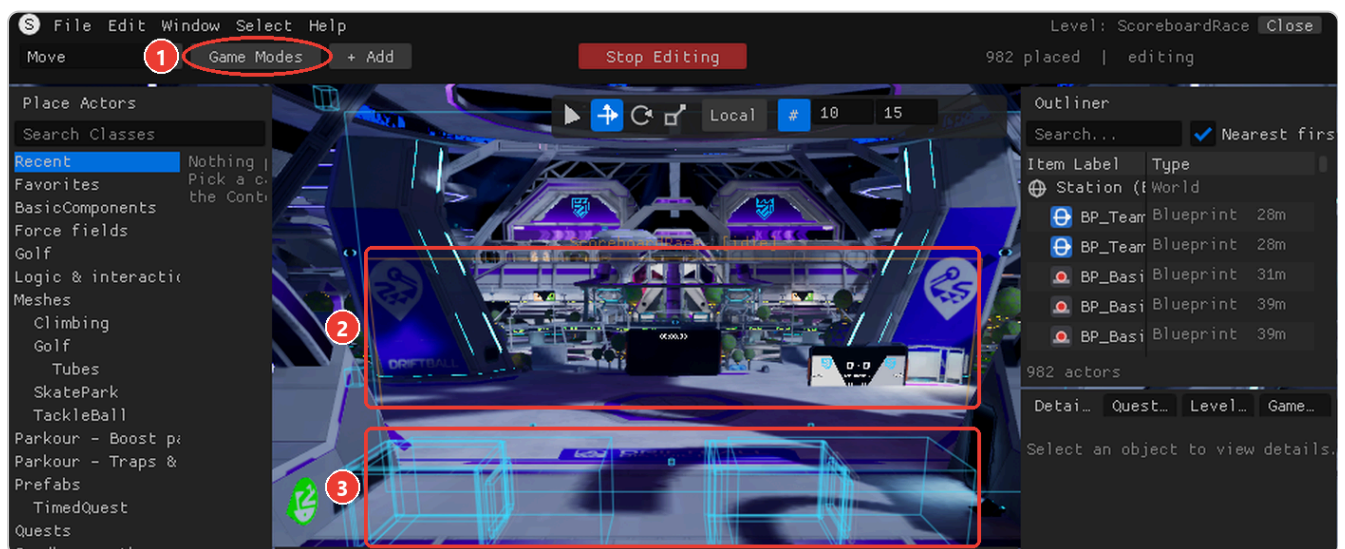
Game modes: `gamemode_create(name, center, teams, team_names, max_players)`, `gamemode_set(mode, key, value)`, `gamemode_control(mode, start|startnow|end|reset)`, `gamemode_team_changer(mode, team, location)`, `gamemode_set_role(mode, handle, role)`, `gamemode_get_code(mode)`, `gamemode_set_code(mode, source)`, `gamemode_apply_script(mode)`, `gamemode_delete(mode)`.

See: `screenshot(max_width)` returns what the editor shows right now as an image -- frame things first with `set_camera`, then `look`. Use it to check placement, facing (scoreboards and signs have a front) and that pieces stand on the floor.

Anything else: `editor_op(line)` runs one editor op (below), `wait(seconds)`.

6 Game modes

A game mode is an area (about 78 x 95 x 19 m, centred where you create it) with **its own teams, rounds and scores**. Players join a team through the mode's **team changers**. The server runs the mode: `idle` -> `countdown` -> `running` -> `ended` -> `idle`.



What a game mode looks like in the editor. **1** the Game Modes window button. **2** the mode's area (orange box) with its name and state. **3** its team changers. The `screenshot` tool shows you the same view.

Rules (gamemode_set): start_mode (manual / button / auto), min_players (per team, for auto), countdown, round_time (0 = no limit), score_to_win (0 = none), end_delay, auto_restart (0/1), stop_when_empty (0/1: end a round when every team is empty -- an empty test arena ends rounds at once), teamN.name, teamN.max, name, and custom.<key> for your own settings.

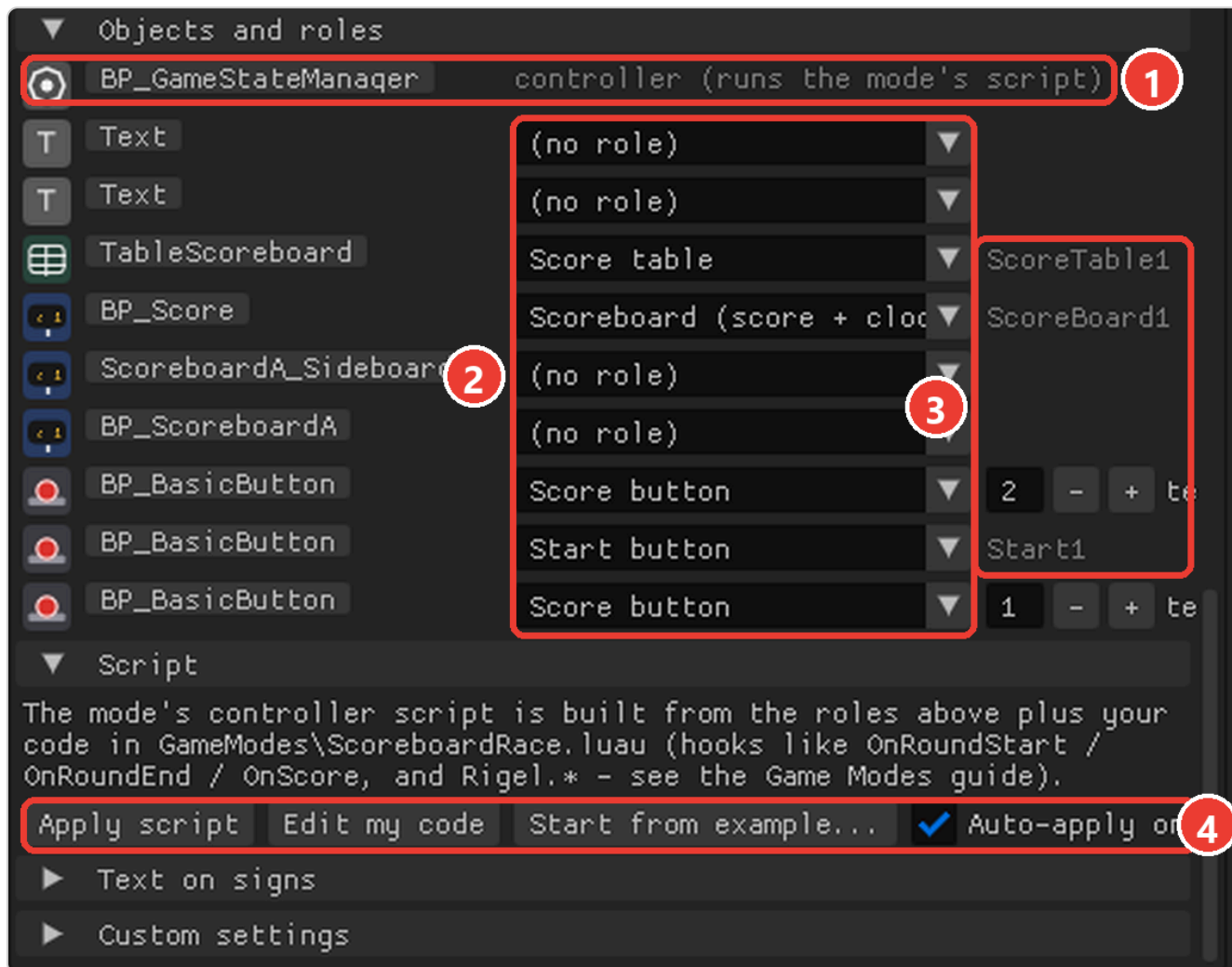
Roles (gamemode_set_role) wire objects inside the mode into its **controller script**, generated from the roles plus the mode's code:

Role	Object	Effect	Slot name in mode code
start	button	starts a round (when none is on)	Start1, Start2, ...
score:N	button	+1 point for team N during a round	Score1, ...
trap_round	trap / toggleable	on during rounds, off otherwise	TrapRound1
trap_pulse:S	trap	flips on/off every S seconds during rounds	TrapPulse1
trap_fired	trap	switched on by trap buttons	TrapFired1
trap_button:S	button	fires every trap_fired trap for S seconds	TrapButton1
wall_lobby	force field / wall	solid and visible between rounds, gone during	WallLobby1
wall_round	force field / wall	only there during rounds	WallRound1

A wall slot is a ToggleableComponent for force fields and shields (on = the wall stands) and a PhysicalComponent (shown / hidden) for anything else.

timer	Timer	counts the countdown, then the round	Timer1
ball	ball spawner	the server spawns / resets its ball each round	Ball1
score_board	the classic Score board	shows team 1 / 2 points	ScoreBoard1
score_table	Score table	a row per team: name, score, players, rounds won	ScoreTable1

The **Scoreboard monitors** (BP_ScoreboardA_C, side and half-court variants) need no role: the controller keeps the mode's team scores and rounds won on them (Gamemode:setTeamScore, setTeamRoundsWon).



The same roles in the Game Modes window: **1** the controller that runs the generated script, **2** each object's role, **3** the slot name the mode code uses, **4** Apply script / Edit my code / Start from example.

Mode code (`gamemode_set_code`) is Luau spliced into the controller. Don't write `BeginPlay`; define hooks:

```
function OnLobby() end -- no round on
function OnCountdown(seconds: number) end -- a round is about to start
function OnRoundStart(round: number) end -- round = 1, 2, ...
function OnRoundEnd(winner: number) end -- winner = team number, 0 = draw
function OnTeamChanged(team: number, size: number, old: number) end
function OnScore(team: number, score: number, old: number) end
function OnTime(secondsLeft: number) end -- every second of a countdown or round
```

The **Rigel library** (in every script): `Rigel.state()` (`idle` / `countdown` / `running` / `ended`), `isRunning()`, `round()`, `timeLeft()`, `teams()`, `teamName(t)`, `teamSize(t)`, `teamMax(t)`, `score(t)`, `roundsWon(t)`, `players()`, `winner()`, `setting(k)`, `settingNumber(k)`; actions `startRound()`, `startNow()`, `endRound(winner?)`, `resetGame()`, `addScore(t, n?)`, `setScore(t, n)`, `setCustom(k, v)`, `setText(name, value)`, `resetBalls()`; callbacks `onStateChanged(f)`, `onTeamChanged(f)`, `onScoreChanged(f)`, `onTimeChanged(f)`. Teams are numbered from **1**.

Text templates for a mode named `Arena`: `{Arena.state}`, `{Arena.time}`, `{Arena.round}`, `{Arena.winner}`, `{Arena.players}`, `{Arena.team1.name}`, `{Arena.team1.score}`, `{Arena.team1.size}`, `{Arena.team1.max}`, `{Arena.team1.wins}`.

Team changer safety. A team changer built on top of a standing player would crash that player's game, so the server builds it beside the player and moves it in when they step away. Deleting the station's own team changers

is refused for the same reason.

Team doors. A team changer's middle is 1.5 m up: the server stands it on the floor under the spot and turns it to the given yaw (players walk through along it). Frame it so players find it -- two cube posts just outside the 3.5 m opening and a beam over the top, in the team's colour, never across the doorway -- and put a Text sign above reading `{<mode>.team1.name} team`. The example levels do exactly this.

Balls. A ball is made at run time; its spawner is what's placed. Moving a ball (transform tool on the ball) moves its spawner by the same amount and the ball comes along; moving the spawner does too. Balls can't be resized: keep ball spawners at scale 1 (the game draws a scaled spawner's ball smaller).

Monitor countdown. Scoreboard monitors show the countdown seconds (round clock waiting) once someone is on a team, then the clock and scores. Keep `countdown` above 0 for a visible count.

Building a game mode, step by step

```
level_new("Duel")
gamemode_create(name="Duel", center=[0,-2600,-27000], teams=2, team_names=["Blue","Red"],
max_players=[4,4])
gamemode_team_changer(mode="Duel", team=1, location=[-800,-6500,-27000])
gamemode_team_changer(mode="Duel", team=2, location=[ 800,-6500,-27000])
place_object("BP_BasicButton_C", [0,-5000,-27000])      -> start button
gamemode_set_role(mode="Duel", handle=<it>, role="start")
place_object("BP_BasicButton_C", [-1200,-2600,-27000])  -> gamemode_set_role(... role="score:1")
place_object("BP_BasicButton_C", [ 1200,-2600,-27000])  -> gamemode_set_role(... role="score:2")
place_object("BP_ScoreboardA_C", [0,-800,-26750])        (no role needed)
place_object("LE_BP_Text_C", [0,-3800,-26750]) -> set_text(<it>, "{Duel.team1.score} :
{Duel.team2.score}")
gamemode_set(mode="Duel", key="start_mode", value="button")
gamemode_set(mode="Duel", key="score_to_win", value="5")
check_lua(source=<code>, gamemode_code=true) -> gamemode_set_code(mode="Duel", source=<code>)
gamemode_control(mode="Duel", action="start") -> wait -> gamemode_list -> script_logs
level_save()
```

7 A complete, more complex script

A capture-the-zone mini game on ordinary objects (not a game mode): a trigger zone scores for whoever stands in it, a button resets, a light shows who owns the zone.

```

--!strict
-- External Dependencies
local Zone: PhysicalComponent = nil -- a floor panel used as the capture zone
local Reset: BasicButtonComponent = nil
local Light: ToggleableComponent = nil
-- End External Dependencies

local holdTime = 10 -- plain variable (not a property)
local owner = 0 -- team that holds the zone
local progress = {} :: { [number]: number } -- seconds held, by team (a cast, not a typed local)

local function say(text: string)
    pcall(function() Gamemode:broadcastEventString("zone", text) end) -- fills {zone} on Text
objects
end

local function capture(team: number)
    owner = team
    say(`Team {team} holds the zone`)
    pcall(function() if team > 0 then Light:luaEnable() else Light:luaDisable() end end)
end

function BeginPlay()
    if (Zone :: PhysicalComponent?) == nil or (Reset :: BasicButtonComponent?) == nil then
        warn("wire Zone and Reset")
        return
    end
    Zone:setTriggerCollision()
    say("Stand in the zone to capture it")
    -- server side: players entering the zone
    Zone.onOverlapByPlayerServer.Listen(function(playerId: number)
        local team = VRPawn.getTeamIndexByID(playerId)
        progress[team] = (progress[team] or 0) + 1
        if progress[team] >= holdTime and owner ~= team then
            capture(team)
        end
    end)
    Reset.onButtonPressEvent.Listen(function()
        progress = {}
        capture(0)
        say("Zone reset")
    end)
    -- a slow decay so a zone has to be defended
    LuauClock.createTimer(5, function()
        for t, v in progress do
            progress[t] = math.max(0, v - 1)
        end
    end)
end
end

```

8 Editor ops (editor_op)

`editor_op(line)` runs one line of the editor's own command language -- the one its test suite uses. Useful ones:

Op	What it does
where	logs the camera position and what it's looking at

<code>camto x y z, camlook pitch yaw</code>	move / aim the editor camera				
<code>pick <Class>, pickat <Class> x y z</code>	make the nearest instance "the last object" for the ops below				
<code>move dx dy dz, rotate yaw, scale sx sy sz, delete</code>	act on the last object				
<code>select, selectadd, dupsel, delsel</code>	the UI's selection, Ctrl+D, Delete				
<code>props [sub.path]</code>	log what Details > Properties shows				
<code>comps</code>	every component of the last object (with collision and TextRender text)				
<code>luaufile <name></code>	attach <code>RigelScripts\<name>.luau</code> to the last object				
<code>`luauref <script> <slot> <Type> self\`</code>	<code><Class> [x y z]</code>	wire a slot			
<code>quest <id> <title...>, qstep, qpub <id> <radiusCm> <timeLimitS> <title...></code>	checkpoint quests: create, add the last object as the next checkpoint, publish				
<code>qcoinrun <seconds> <coins> <title...></code>	publish a red-coin run				
<code>coinrun <sec> <questHex32></code>	a red-coin run 4 m ahead				
<code>`scene save\`</code>	<code>saveas\`</code>	<code>open\`</code>	<code>close\`</code>	<code>upload [name]\`</code>	levels (the <code>level_*</code> tools use these)
<code>undo, redo</code>	Ctrl+Z / Ctrl+Y				
<code>slots</code>	every module area as this client sees it				
<code>uitour</code>	open every tab and popup in turn (a UI self-test)				

9 When something goes wrong

- **"No level is open":** `level_new(name)` or `level_open(name)` first.
- **Placed but nothing appeared:** the server refused it -- read `editor_problems` and `editor_logs(filter="SPECEDIT")`. Team changers only work inside a game mode. Some prefabs can't be placed (the palette's `note` says why).
- **The script never logs:** `script_logs(script="Name")`. No lines at all usually means it failed to compile: look for `LogLuau: Error` lines. Did you `check_luau` it?
- `attempt to index nil`: an unwired slot, or a component the object doesn't have. Nil-check slots; confirm the target with `find_reference_targets`.

- **Works for you, not for others:** you used client-only state (the local player, a local variable) for something shared. Move it to server events / `Rigel` actions.
- **The whole round ends instantly in tests:** `stop_when_empty` is on and nobody is in the teams.
- **The editor isn't running:** every tool fails with "The Spec Editor isn't running"; start it and join a server.